

A LINEARISATION USING PERTURBATION THEORY

In this appendix, we flesh out the application of perturbation theory that leads from the Taylor series expansion to the zeroth-order and first-order systems of equations.

A.1 Recalled equations

The governing RANS momentum equation is

$$U_j \frac{\partial U_i}{\partial x_j} = \frac{\partial}{\partial x_j} K \left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i} \right) - \frac{\partial P}{\partial x_i} + f_i, \quad (\text{A.1})$$

subject to the continuity equation

$$\frac{\partial U_i}{\partial x_i} = 0, \quad (\text{A.2})$$

and boundary conditions

$$\begin{aligned} U_i(z_0) &= 0, \\ U_i(z_i) &= U_i^{\text{lid}}. \end{aligned} \quad (\text{A.3})$$

The velocity and pressure fields are expanded as a perturbation series in the small parameter ε :

$$\begin{aligned} U_i &= U_i^0 + \varepsilon u_i^1 + \varepsilon^2 u_i^2 + \varepsilon^3 u_i^3 + \dots \\ P &= P^0 + \varepsilon p^1 + \varepsilon^2 p^2 + \varepsilon^3 p^3 + \dots \end{aligned} \quad (\text{A.4})$$

A.2 Derivation of the zeroth-order equations

We start by substituting eq. (A.4) into the momentum equation (A.1) and the continuity equation (A.2) to yield the expanded momentum equation,

$$\begin{aligned} & (U_j^0 + \varepsilon u_j^1 + \varepsilon^2 u_j^2 + \varepsilon^3 u_j^3 + \dots) \frac{\partial}{\partial x_j} (U_i^0 + \varepsilon u_i^1 + \varepsilon^2 u_i^2 + \varepsilon^3 u_i^3 + \dots) \\ &= \frac{\partial}{\partial x_j} K \left(\frac{\partial}{\partial x_j} (U_i^0 + \varepsilon u_i^1 + \varepsilon^2 u_i^2 + \varepsilon^3 u_i^3 + \dots) \right. \\ & \quad \left. + \frac{\partial}{\partial x_i} (U_j^0 + \varepsilon u_j^1 + \varepsilon^2 u_j^2 + \varepsilon^3 u_j^3 + \dots) \right) \\ & \quad - \frac{\partial}{\partial x_i} (P^0 + \varepsilon p^1 + \varepsilon^2 p^2 + \varepsilon^3 p^3 + \dots) + f_i, \end{aligned} \quad (\text{A.5})$$

and the expanded continuity equation,

$$\frac{\partial}{\partial x_i} (U_i^0 + \varepsilon u_i^1 + \varepsilon^2 u_i^2 + \varepsilon^3 u_i^3 + \dots) = 0. \quad (\text{A.6})$$

The n th-order equations are obtained by applying the operator $\frac{\partial^n}{\partial \varepsilon^n}$ to both sides and setting $\varepsilon = 0$. To derive the zeroth-order equations, we apply $\frac{\partial^0}{\partial \varepsilon^0}$ to equation (A.5), which does not change the equation, and then set $\varepsilon = 0$ to give

$$U_j^0 \frac{\partial U_i^0}{\partial x_j} = \frac{\partial}{\partial x_j} K \left(\frac{\partial U_i^0}{\partial x_j} + \frac{\partial U_j^0}{\partial x_i} \right) - \frac{\partial P^0}{\partial x_i}. \quad (\text{A.7})$$

The continuity equation and the boundary conditions (A.3) remain unchanged, giving the complete zeroth-order system:

$$\begin{aligned} U_j^0 \frac{\partial U_i^0}{\partial x_j} &= \frac{\partial}{\partial x_j} K \left(\frac{\partial U_i^0}{\partial x_j} + \frac{\partial U_j^0}{\partial x_i} \right) - \frac{\partial P^0}{\partial x_i} \\ \frac{\partial U_i^0}{\partial x_i} &= 0 \\ U_i^0(z_0) &= 0 \\ U_i^0(z_i) &= U_i^{\text{lid}}. \end{aligned} \quad (\text{A.8})$$

Note that U_i^{lid} is an externally prescribed quantity and therefore not a function of ε .

A.3 Derivation of the first-order equations

The first-order equations are derived by applying the operator $\frac{\partial}{\partial \varepsilon}$ to equation (A.5). Starting with the convective term, we apply the product rule to find

$$\begin{aligned} \frac{\partial}{\partial \varepsilon} \left[(U_j^0 + \varepsilon u_j^1 + \dots) \frac{\partial}{\partial x_j} (U_i^0 + \varepsilon u_i^1 + \dots) \right] \\ = (U_j^0 + \varepsilon u_j^1 + \dots) \frac{\partial}{\partial x_j} (u_i^1 + \dots) \\ + (u_j^1 + \dots) \frac{\partial}{\partial x_j} (U_i^0 + \varepsilon u_i^1 + \dots). \end{aligned} \quad (\text{A.9})$$

Setting $\varepsilon = 0$, you recover

$$\frac{\partial}{\partial \varepsilon} \left[(U_j^0 + \varepsilon u_j^1 + \dots) \frac{\partial}{\partial x_j} (U_i^0 + \varepsilon u_i^1 + \dots) \right] \Big|_{\varepsilon=0} = U_j^0 \frac{\partial u_i^1}{\partial x_j} + u_j^1 \frac{\partial U_i^0}{\partial x_j}. \quad (\text{A.10})$$

Taking the derivative with respect to ε of the Reynolds stress term, and subsequently setting $\varepsilon = 0$, we get

$$\begin{aligned} \frac{\partial}{\partial \varepsilon} \left[\frac{\partial}{\partial x_j} K \left(\frac{\partial}{\partial x_j} (U_i^0 + \varepsilon u_i^1 + \dots) + \frac{\partial}{\partial x_i} (U_j^0 + \varepsilon u_j^1 + \dots) \right) \right] \Big|_{\varepsilon=0} \\ = \frac{\partial}{\partial x_j} K \left(\frac{\partial u_i^1}{\partial x_j} + \frac{\partial u_j^1}{\partial x_i} \right). \end{aligned} \quad (\text{A.11})$$

Following the same procedure with the pressure term, we obtain

$$\frac{\partial}{\partial \varepsilon} \left[-\frac{\partial}{\partial x_i} (P^0 + \varepsilon p^1 + \dots) \right] \Big|_{\varepsilon=0} = -\frac{\partial p^1}{\partial x_i}. \quad (\text{A.12})$$

Finally, the forcing term is simply

$$\frac{\partial}{\partial \varepsilon} (\varepsilon f_i) \Big|_{\varepsilon=0} = f_i. \quad (\text{A.13})$$

Collecting these terms together, we arrive at the following first-order momentum equation

$$U_j^0 \frac{\partial u_i^1}{\partial x_j} + u_j^1 \frac{\partial U_i^0}{\partial x_j} = \frac{\partial}{\partial x_j} K \left(\frac{\partial u_i^1}{\partial x_j} + \frac{\partial u_j^1}{\partial x_i} \right) - \frac{\partial p^1}{\partial x_i} + f_i. \quad (\text{A.14})$$

A.4 Simplification for unidirectional shear flow

Equation (A.14) is the general first-order momentum equation. To arrive at the simplified form used in the model, we make two additional assumptions about the approaching flow: (i) it is along the x -axis, and (ii) it is only a function of the vertical coordinate, z , such that

$$U_i^0 = \delta_{i1} U^0(z) = \begin{pmatrix} U^0(z) \\ 0 \\ 0 \end{pmatrix}, \quad (\text{A.15})$$

where δ_{i1} is the Kronecker delta function. This impacts only the terms on the left-hand side of (A.14). Substituting in the first term, we can replace U_j^0 with $\delta_{j1} U^0(z)$ to obtain

$$\begin{aligned} U_j^0 \frac{\partial u_i^1}{\partial x_j} &= \delta_{j1} U^0(z) \frac{\partial u_i^1}{\partial x_j} \\ &= \delta_{11} U^0(z) \frac{\partial u_i^1}{\partial x} + \delta_{12} U^0(z) \frac{\partial u_i^1}{\partial y} + \delta_{13} U^0(z) \frac{\partial u_i^1}{\partial z} \\ &= U^0(z) \frac{\partial u_i^1}{\partial x}. \end{aligned} \quad (\text{A.16})$$

The second term is, analogously,

$$\begin{aligned}
u_j^1 \frac{\partial U_i^0}{\partial x_j} &= u_j^1 \frac{\partial}{\partial x_j} (\delta_{il} U^0(z)) \\
&= u^1 \frac{\partial}{\partial x} \delta_{il} U^0(z) + v^1 \frac{\partial}{\partial y} \delta_{il} U^0(z) + w^1 \frac{\partial}{\partial z} \delta_{il} U^0(z) \\
&= w^1 \frac{\partial U^0}{\partial z} \delta_{il}.
\end{aligned} \tag{A.17}$$

Introducing eqs. (A.16) and (A.17) into eq. (A.14) gives the simplified first-order system:

$$U^0 \frac{\partial u_i^1}{\partial x} + w^1 \frac{\partial U^0}{\partial z} \delta_{il} = \frac{\partial}{\partial x_j} K \left(\frac{\partial u_i^1}{\partial x_j} + \frac{\partial u_j^1}{\partial x_i} \right) - \frac{\partial p^1}{\partial x_i} + f_i \tag{A.18a}$$

$$\frac{\partial u_i^1}{\partial x_i} = 0 \tag{A.18b}$$

$$u_i^1(z_0) = 0 \tag{A.18c}$$

$$u_i^1(z_i) = 0. \tag{A.18d}$$

Finally, the first-order continuity equation is derived as

$$\frac{\partial}{\partial \varepsilon} \left[\frac{\partial}{\partial x_i} (U_i + \varepsilon u_i^1 + \dots) \right] \Big|_{\varepsilon=0} = \frac{\partial u_i^1}{\partial x_i} = 0, \tag{A.19}$$

and the boundary conditions are given as

$$\frac{\partial}{\partial \varepsilon} (U_i^0 + \varepsilon u_i^1 + \dots) \Big|_{\varepsilon=0} \Big|_{z_0} = 0 \quad \Rightarrow \quad u_i^1(z_0) = 0, \tag{A.20}$$

at roughness height z_0 , and

$$\frac{\partial}{\partial \varepsilon} (U_i^0 + \varepsilon u_i^1 + \dots) \Big|_{\varepsilon=0} \Big|_{z_i} = \frac{\partial U_i^{\text{lid}}}{\partial \varepsilon} \quad \Rightarrow \quad u_i^1(z_i) = 0 \tag{A.21}$$

at inversion height z_i .